

Cloudsploitable File Server

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Changes since last presentation

- ♦ Rebranded the product to Cloudsploitable
- ♦ Improved browser UI for login, file listing, upload, and download
- ♦ Added automated JUnit tests and GitHub Actions CI/CD
- ♦ Added /oe product metrics endpoint so the OE dashboard can pull product data, including customer disruption alert metrics
- ♦ Added additional supporting analysis documents as required by weekly homework

Improved UI

Cloudsploitable: A Vulnerable Cloud Server

Server: **Online** User: **None** Token: **Missing**

Login

Refresh Files

Upload

Download

Files Available

Login is required first

File Name	Size	Uploaded At	Uploaded By

Downloaded File Content

Improved UI

Cloudsploitable: A Vulnerable Cloud Server

Server: **Online** User: **admin** Token: **Present**

Logged In

Refresh Files

Upload

Download

Download File

File Name

readme.txt

Submit Download

Files Available

File list refreshed

File Name	Size	Uploaded At	Uploaded By
notes.txt	38 B	2026-04-19 14:37:08	test
hello.txt	33 B	2026-04-22 11:05:41	admin
readme.txt	29 B	2026-04-18 09:14:22	haomin
really-long-file.txt	3.4 KB	2026-05-12 09:42:46	test

Downloaded File Content

Top three business risks

- ♦ Unauthorized access or data exposure
Exposing sensitive customer data could result in loss of trust, reputational damage, and legal or regulatory concerns.
- ♦ Service disruption or degraded core functionality
A prolonged inability to log in or use key product features may cause customers to switch to another similar service.
- ♦ Platform misuse and accountability risk
If customers use the service to store harmful or illegal content, the business may face legal concerns, reputational damage, or investigation costs.

Updated OE Dashboard

- ◆ Now includes a customer disruption alert metrics table

Customer Disruption Alert Metrics

Risk Area	Status	Metric	Current Value	Alert Rule
Login disruption	OK	Valid Login Success Rate	92%	OK >= 90%, Warn 75-89%, Critical < 75%
Session access disruption	WARN	Protected Endpoint Access Success Rate	85%	OK >= 90%, Warn 75-89%, Critical < 75%
File operation disruption	WARN	File Operation Success Rate	81%	OK >= 90%, Warn 75-89%, Critical < 75%

Adjustments made

- ♦ Added /oe endpoint to the Java product

```
server.createContext(path: "/oe", new OeHandler(authManager, fileStorage));
```

- ♦ Returns some live data, while others are simulated

```
return Map.of(
    k1: "lastUpdated", LocalDateTime.now().format(DateTimeFormatter.ofPattern("yyyy-MM-dd HH:mm:ss")),
    k2: "healthStatus", v2: "Healthy",
    k3: "healthSubtext", v3: "GET /oe returned 200 OK",
    k4: "avgResponseTime", v4: "118 ms",
    k5: "storageUsed", v5: storagePercent + "%",
    k6: "storageSubtext", v6: "Used: " + formatBytes(storageUsedBytes) + " of " + formatBytes(SIMULATED_STORAGE_SIZE),
    k7: "activeSessions", authManager.getActiveSessionCount(),
    k8: "blockedUsers", authManager.getBlockedUserCount()
);
```